

Thomas GARABEDIAN

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PROFILE

Engineering student at Télécom SudParis, joining the **M2 Data Science** program at École Polytechnique as a double degree. Interested in **AI, machine learning** and **computer vision**, especially for **defense, aerospace, autonomous systems** and **predictive maintenance** applications.

EDUCATION

Institut Polytechnique de Paris, École Polytechnique Sep. 2026 – 2027
Master 2 Data Science (M2DS)

- Program focused on statistical learning, deep learning, generative models, computer vision and modern AI.
- Relevant coursework: **Machine Learning, Deep Learning, Computer Vision, Graph Neural Networks, Large Language Models, Generative Models, Machine Learning, Optimization for data science, model training and deployment**
- Academic objective: strengthen skills in modeling, evaluation, generalization, calibration, interpretability and industrialization of AI prototypes.

Institut Polytechnique de Paris, Télécom SudParis 2024 – 2027
Engineering Degree in Digital Technologies

- **GPA: 3.81/4.0**, top 15% among 220 students.
- Relevant coursework: **Applied Statistics, Optimization, Signal Processing, Data Mining, Data Analysis, Stochastic Processes**, cryptography, secure communications.
- Broad foundation in applied mathematics, computer science, telecommunications, signal processing, cybersecurity and digital systems.

CPGE Lycée Thiers & Prytanée National Militaire 2021 – 2024
Intensive preparatory classes in mathematics, physics and chemistry (PCSI then PC)

- Highly selective program preparing for French engineering school entrance examinations.
- Developed strong analytical rigor, work capacity and problem-solving methods under time constraints.

Lycée Lacordaire 2020 – 2021
French Baccalaureate, Highest Honors

AI & DATA SCIENCE PROJECTS

HAKS Hackathon, Airbus x IBM | 1st Prize, 2,500 € Jun. 2026
Télécom Paris *Palaiseau, France*

- Developed a **probabilistic corrosion risk prediction model** for a given inspection date, using operational aerospace data.
- Leveraged domain variables: aircraft age, environmental exposure, ground/flight ratio, C-checks, corrosion history and inspection data.
- Combined advanced tabular models, including **FT-Transformer** and **XGBoost**, with a focus on probability calibration and predictive performance.
- Evaluated with the **Brier Score**; achieved **0.150**. Final use case: prioritize inspections, support maintenance decisions and reduce operational risk.

D4GEN Hackathon, Genopole x AWS | 1st Prize, 10,000 € May – Jun. 2026
Genopole d'Évry *Évry, France*

- Designed a **human-in-the-loop** tool to accelerate expert annotation for animal species monitoring using weakly labeled data.
- Built a pipeline combining **unsupervised clustering**, clip export, spectrograms and an annotation API.
- Fine-tuned **SAM 3** for segmentation and visual identification support, aiming to reduce annotation time and improve label quality.
- Worked through rapid iterations with expert users, feature prioritization and a technical presentation to the jury.

PROFESSIONAL EXPERIENCE

CIEDS IP Paris Project | Drone Swarm Automation *ENSTA Paris*

Jan. – Jun. 2026
Palaiseau, France

- Designed a **computer vision** pipeline to detect vehicles and buildings in aerial images within a multi-drone surveillance scenario.
- Compared object detection models: **CNN, YOLO, DETR**. Analyzed trade-offs between accuracy, robustness, computation cost and operational integration.
- Contributed to swarm coordination: target assignment, trajectory planning, scenario simulation and decision logic structuring.
- Presented results to **CIEDS IP Paris**, highlighting model trust and defense-oriented deployment issues.

Engineering Internship | ALSEAMAR, Alcen Group *Embedded systems, RF and industrial software*

Jun. – Aug. 2025
Six-Fours, France

- Developed control software for a **high-frequency power amplifier**, used in an industrial environment and for submarine antenna testing.
- Implemented modules for equipment communication, supervision, acquisition, data logging, diagnostics and measurement traceability.
- Carried out validation tests to improve system reliability, reduce user errors and strengthen experimental measurement tracking.
- Worked at the interface of **Python**, embedded systems, RF electronics, signal processing and industrial constraints in a French defense company.

Tennis Coach *Sports instruction, pedagogy and group management*

2019 – 2024
Marseille, France

- Organized and led tennis camps for groups of up to 30 participants; adapted drills to age, level and sporting objectives.
- Provided individual coaching focused on technical, tactical and mental preparation for regional competitions; developed pedagogy and field leadership skills.

Astronomy Mission | Saint-Véran Observatory *Scientific observation and data analysis*

Summer 2022
Saint-Véran, France

- Searched and analyzed astronomical data for international researchers.
- Practiced astrophysics methods: acquisition, verification, interpretation and documentation of measurements; introduction to photometry, astrometry and spectrometry.

Observation Internships | Société Générale and Cabinet Hermessiane *Finance, accounting and digital strategy*

Summer 2017 – Summer 2018
Paris, France

- Observed directors, engineers, financial analysts and accounting professionals; learned about monitoring, analysis and coordination methods.
- Attended client meetings and year-end accounting reviews; contributed to a reflection on a company's digital strategy.

SKILLS

AI & Computer Vision

object detection, segmentation, classification, CNN, YOLO, DETR, SAM 3, Transformers, fine-tuning, human-in-the-loop annotation, aerial image analysis, spectrograms.

Machine Learning

FT-Transformer, XGBoost, LightGBM, CatBoost, unsupervised clustering, tabular models, probability calibration, error analysis, feature engineering, model experimentation.

Model Evaluation

IoU, precision, recall, Brier Score, WAPE, classification and segmentation metrics, robustness, interpretability, performance / compute-cost trade-offs.

Programming & Tools

Python, experimental pipelines, annotation APIs, data acquisition and logging, diagnostic tools, dataset structuring, technical reporting.

Industrial Systems

embedded systems, RF electronics, signal processing, equipment supervision, software validation, measurement traceability, maintenance and field-use constraints.

Soft Skills

leadership, pedagogy, decision-making under constraints, technical communication, teamwork, short-cycle project management, oral presentations to juries and stakeholders.

LEADERSHIP & ENGAGEMENT

Corporate Relations Manager | Lib'Ensemble

2024 – 2025

Student association with social impact

Évry, France

- Searched for funding, medicine donations and partnerships to support aid shipments to Zahlé, Lebanon; raised more than 3,000 €.
- Contacted companies and partners, prepared discussions, tracked contributions and coordinated with the student team.

Volunteer Work in a Nursing Home and a Medical-Educational Institute

2019 – 2021

Social support and activity organization

Marseille, France

- Organized activities and academic support; contributed to the well-being and development of residents and young people.
- Developed interpersonal skills: active listening, patience, communication adaptation and sense of responsibility.

Founder of TedLaco

2020 – 2021

Conference series on science, society and politics

Lycée Lacordaire, Marseille

- Created and hosted a conference series on cross-disciplinary topics linking science, politics and society.
- Project awarded the **Grand Prix de l'Innovation** at Lycée Lacordaire.

CERTIFICATIONS & AWARDS

- **Special Military Preparation** (2023): intensive training program in military fundamentals, leadership and decision-making under constraints, École du Génie de Thorée-les-Pins.
- **Model United Nations Certificate** (2017, 2018): active participation over two consecutive years; diplomacy, negotiation, argumentation and public speaking.
- **Projet Voltaire, Business Level** (2020): advanced certification in French language proficiency, with professional and business specialization.
- **Tennis Referee Diploma** (2019): regional first prize, top of cohort; command of tennis regulations and competition situations.

LANGUAGES

- **French:** native.
- **English:** C1.
- **Spanish:** C1.
- **Italian:** beginner.

INTERESTS

- **Sports:** competitive tennis, tennis refereeing, hiking.
- **Science & technical culture:** science communication, science news, cosmology, astrophysics, military and warfare history.
- **Professional interests:** defense, aerospace, predictive maintenance, autonomous systems, embedded AI, model safety and interpretability.